

EXHIBIT 17

Ex. 17 – Infringement Contention Claim Chart
U.S. Patent No. 12,530,925 (“the ’925 patent”)

Apple’s products directly and/or indirectly infringe at least claims 1, 5, 6, 7, 8, 17, 21, 22, 23, 24 (“Asserted Claims”) of U.S. Patent No. 12,530,925 (“the ’925 patent”). The products that infringe the ’925 patent include at least the iPhone 15, iPhone 15 Plus, iPhone 15 Pro, iPhone 15 Pro Max, iPhone 16, iPhone 16e, iPhone 16 Pro, iPhone 16 Pro Max, iPhone 16 Plus, iPhone 17, iPhone 17e, iPhone 17 Pro, iPhone 17 Pro Max, iPhone Air, 11-inch iPad Pro (4th Generation), iPad Pro 12.9 inch (6th Generation), iPad Pro 11-inch (M4 and M5), iPad Pro 13-inch (M4 and M5) and other infringing iPhone and iPad models (each an “Accused Apple Product” and collectively the “Accused Apple Products”). The Accused Apple Products infringe the ’925 patent as set forth below. Each of the following limitations is met literally, and, to the extent a limitation is not met literally, it is met under the doctrine of equivalents.

Additionally, the Apple iPhone 17 is representative of the Accused Apple Products at least because the other Accused Apple Products function in the same or substantially similar manner to the Apple iPhone 17.¹ The relevant identification processes and functionalities, as well as the hardware components of the Apple iPhone 17, are demonstrated through testing of the Apple iPhone 17 (“Apple Product Testing”), which is available for inspection upon request.

Claim 1	Accused Apple Products
[1pre] A system comprising:	The Accused Apple Products comprise a system. For example, the Apple iPhone 17 is a system:

¹ Each of the Accused Apple Products includes the claimed hardware components and steps. The hardware components and source code implementing the relevant functionalities of the Apple iPhone 17 are representative of the Accused Apple Products for the purpose of infringement. To the extent there may be differences in the hardware composition, certain software steps, or source code versions among these Accused Apple Products, those differences are not material for the infringement analysis.

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Exhibit 36.

[1a] a near infrared (NIR) camera configured to capture a scene, the NIR camera comprising a CMOS sensor;

The Accused Apple Products comprise a near infrared (NIR) camera configured to capture a scene, the NIR camera comprising a CMOS sensor.

“After the TrueDepth camera confirms the presence of an attentive face, it projects and reads thousands of infrared dots to form a depth map of the face along with a 2D infrared image.”
Exhibit 35, at 22.

The Apple iPhone 17 Face ID module includes a near-infrared camera module.

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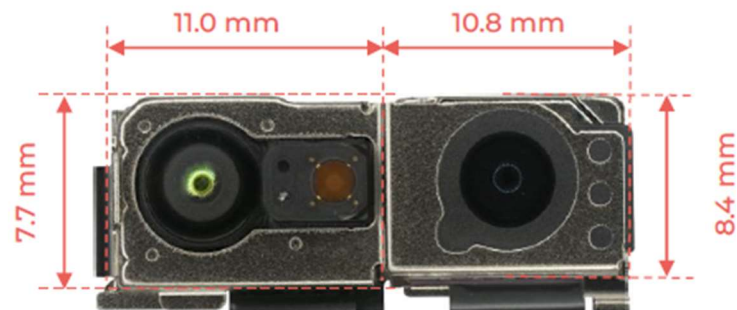
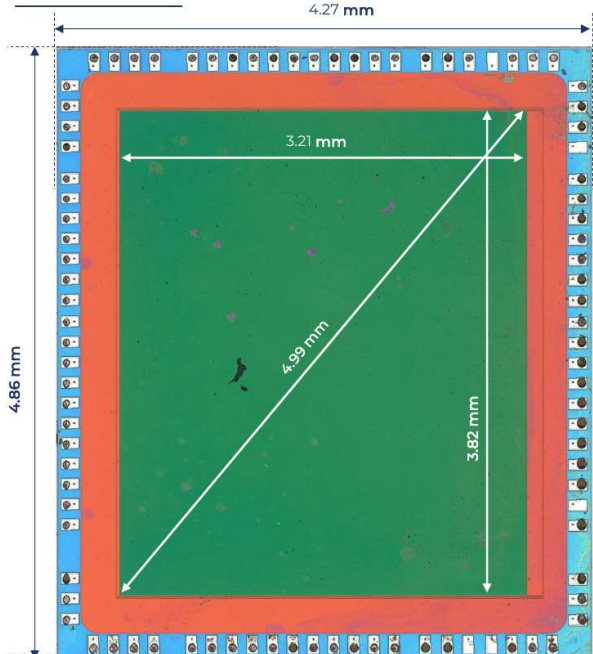
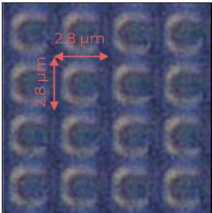


Exhibit 34C, at 26.

The near-infrared camera uses a monochrome NIR CMOS image sensor.

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	NIR CMOS IMAGE SENSOR DIE Overview  - Die area: 20.75 mm² (4.86 x 4.27 mm) - Nb of PGDW per 12-inch wafer: 3,076 - Pad number: 94 - Pixel array area: 12.26 mm² (3.82 x 3.21 mm) - Pixel size: 7.84 μm² (2.8 x 2.8 μm) - Resolution: 1.56 Mp (1,364 x 1,146) - Color Filter: Monochrome - Pattern: - Marking: S6G812A1A   ipugan@trinamix.de CIS Die Overview - Optical View ©2026 Yole Group SPR26034 Apple iPhone 17 Face ID www.yolegroup.com 46 Exhibit 34C, at 46.
[1b] an illumination unit comprising at least one of an LED and a laser, the illumination unit configured to project NIR light in a light pattern toward the scene;	The Accused Apple Products comprise an illumination unit comprising at least one of an LED and a laser, the illumination unit configured to project NIR light in a light pattern toward the scene. The TrueDepth camera system includes a dot projector Vertical-Cavity Surface-Emitting Laser (“VCSEL”), a flood illumination VCSEL, and a near-infrared camera module.

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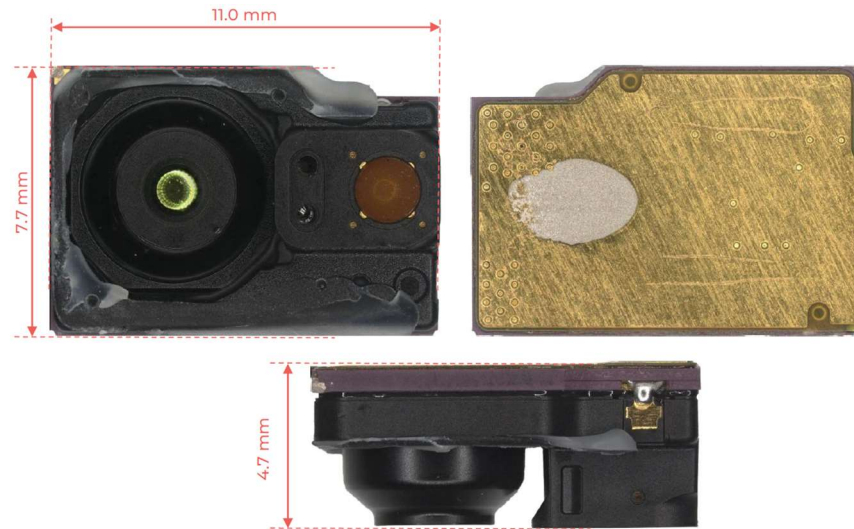


Exhibit 34C, at 35.

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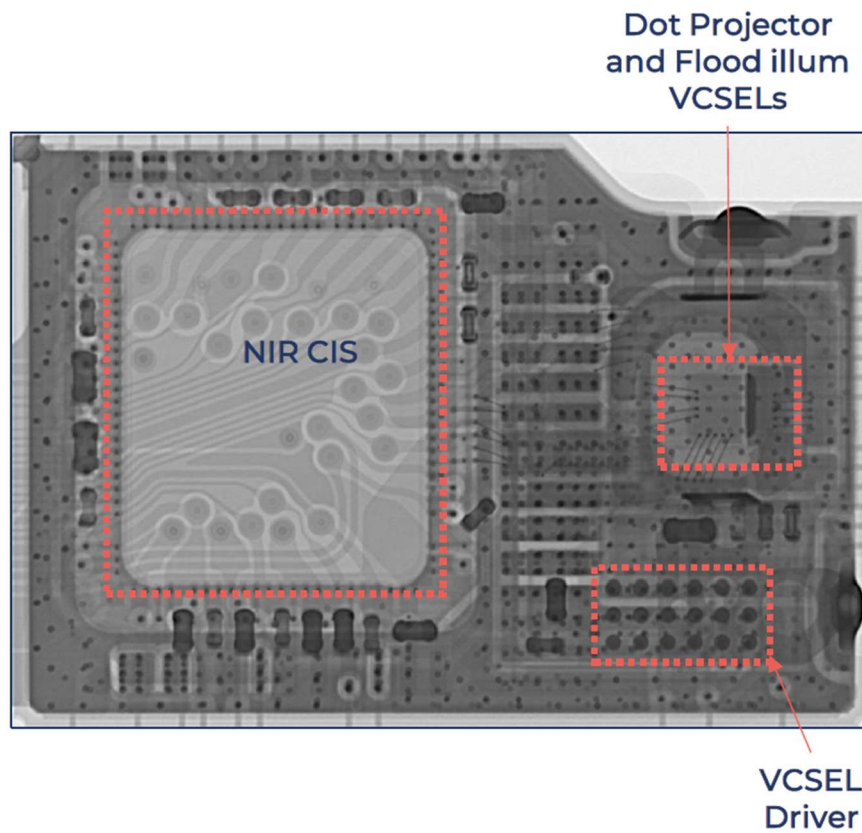
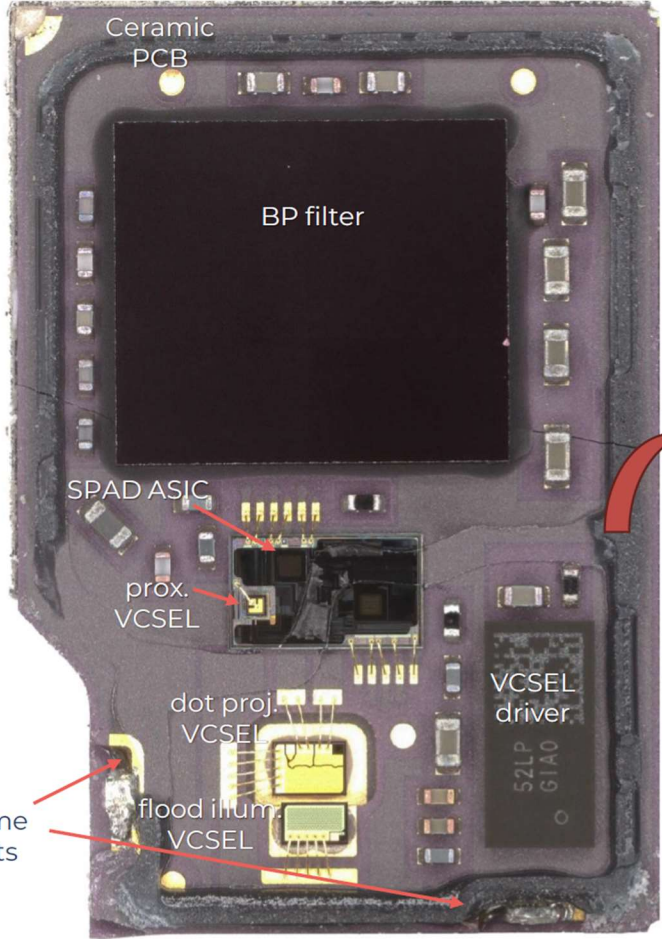


Exhibit 34C, at 36.

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	 The photograph shows the internal components of an iPhone 17 Pro Face ID module. Labels with red arrows point to the following parts: Ceramic PCB (top), BP filter (large black square), SPAD ASIC (small chip), prox. VCSEL (small chip), dot proj. VCSEL (small chip), flood illum. VCSEL (small chip), DOE leadframe contacts (bottom left), and VCSEL driver (chip labeled 52LP GIA0). A red curved arrow points to the right side of the module. iPhone 17 Pro Face ID Module Opened © Yole Group 2026 Exhibit 34C, at 37.
[1c] at least one optical element, wherein the NIR light passes	The Accused Apple Products comprise at least one optical element, wherein the NIR light passes through the at least one optical element.

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through the at least one optical element;

The iPhone 17 comprises a diffractive optical element the NIR light passes through:

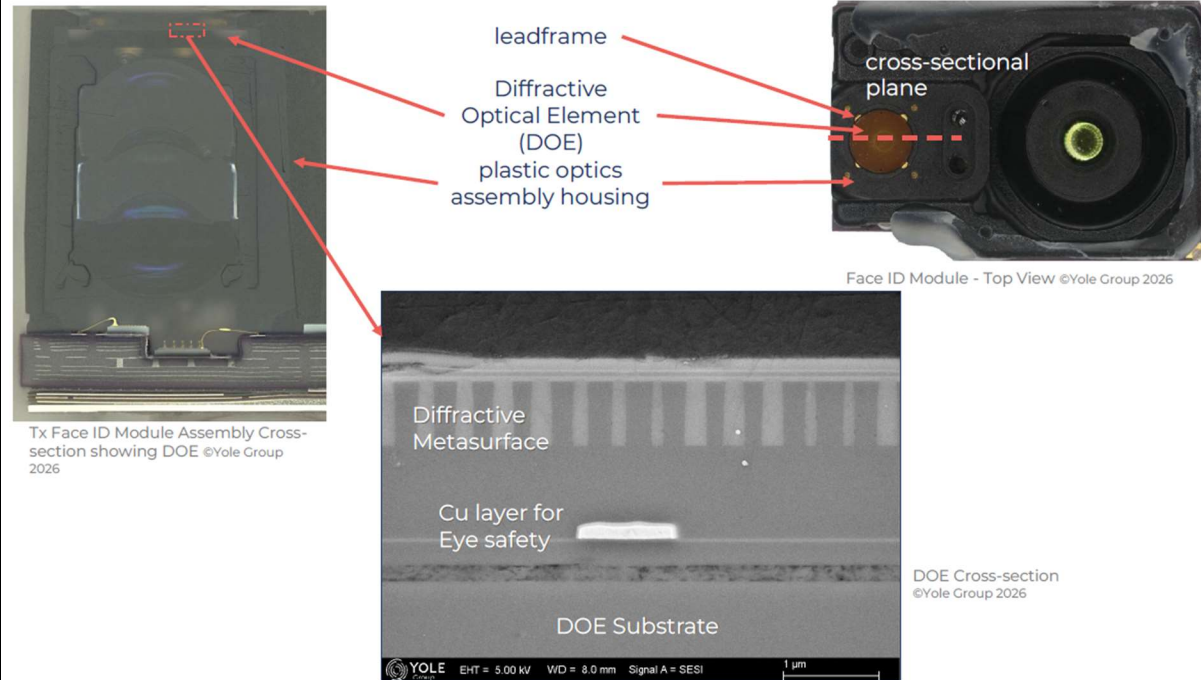
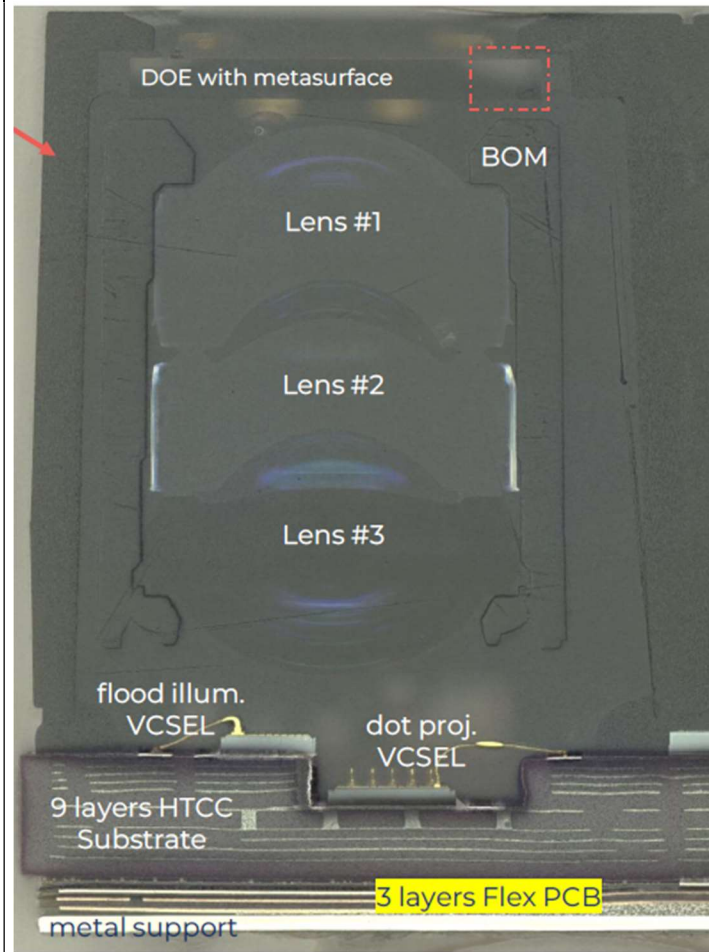


Exhibit 34C, at 66.

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Cross Section of iPhone 17 Face ID (TX) – SEM View

Exhibit 34C, at 69.

[1d] a processor configured to:

The Accused Apple Products comprise a processor.

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	For example, the Apple iPhone 17 includes a processing chip. Chip  A19 chip 6-core CPU with 2 performance and 4 efficiency cores 5-core GPU with Neural Accelerators 16-core Neural Engine Hardware-accelerated ray tracing Exhibit 33.
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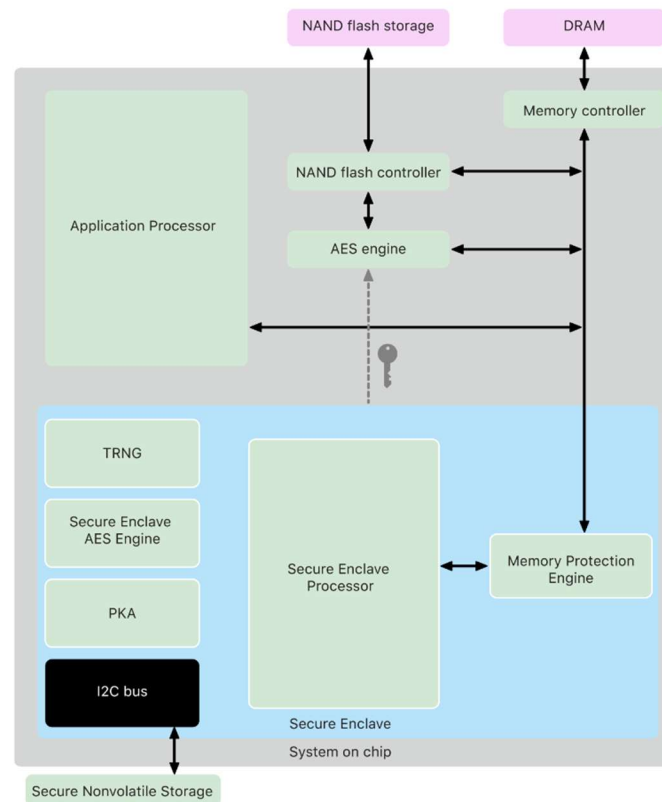
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Secure Enclave

The Secure Enclave is a dedicated secure subsystem in the latest versions of iPhone, iPad, Mac, Apple TV, Apple Watch, Apple Vision Pro, and HomePod.

Overview

The Secure Enclave is a dedicated secure subsystem integrated into Apple [system on a chip \(SoC\)](#). The Secure Enclave is isolated from the main processor to provide an extra layer of security and is designed to keep sensitive user data secure even when the Application Processor kernel becomes compromised. It follows the same design principles as the SoC does—a boot ROM to establish a hardware root of trust, an AES engine for efficient and secure cryptographic operations, and protected memory. Although the Secure Enclave doesn’t include storage, it has a mechanism to store information securely on attached storage separate from the NAND flash storage that’s used by the Application Processor and operating system.



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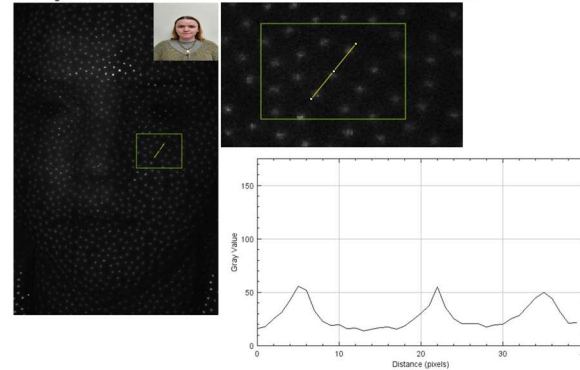
	Exhibit 35, at 10.
[1e] output a first image comprising light reflected from the scene and detect a face in the first image;	The Accused Apple Products comprise a processor configured to output a first image comprising light reflected from the scene and detect a face in the first image. “Face ID is designed to confirm user attention, provide robust authentication with a low false-match rate, and mitigate both digital and physical spoofing.” Exhibit 35, at 22. “The TrueDepth camera automatically looks for the user’s face when the user wakes an Apple device that features Face ID (by raising it or tapping the screen), as well as when those devices attempt to authenticate the user to display an incoming notification or when a supported app requests Face ID authentication. When a face is detected, Face ID confirms attention and intent to unlock by detecting that the user’s eyes are open and their attention is directed at their device.” Exhibit 35, at 22. “Facial matching is performed within the Secure Enclave using neural networks trained specifically for that purpose. Apple developed the facial matching neural networks using over a billion images, including infrared (IR) and depth images collected in studies conducted with the participants’ informed consent. Apple then worked with participants from around the world to include a representative group of people accounting for gender, age, ethnicity, and other factors. The studies were augmented as needed to provide a high degree of accuracy for a diverse range of users. Face ID is designed to work with hats, scarves, eyeglasses, contact lenses, and many types of sunglasses. Face ID also supports unlocking with a mask on iPhone devices starting with iPhone 12 and iOS 15.4 or later.” Exhibit 35, at 29.
[1f] output a second image comprising a reflective light pattern;	The Accused Apple Products comprise a processor configured to output a second image comprising a reflective light pattern. “After the TrueDepth camera confirms the presence of an attentive face, it projects and

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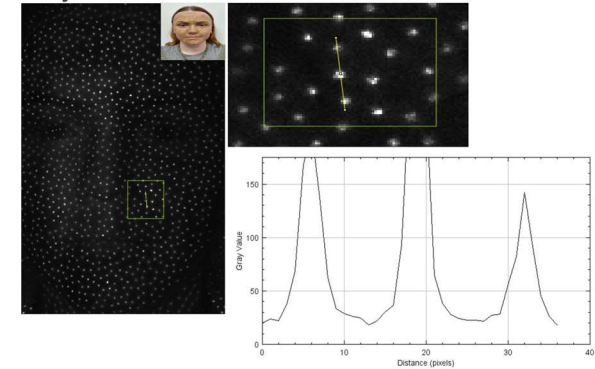
reads thousands of infrared dots to form a depth map of the face along with a 2D infrared image. This data is used to create a sequence of 2D images and depth maps, which are digitally signed and sent to the Secure Enclave. To counter both digital and physical spoofs, the TrueDepth camera randomizes the sequence of 2D images and depth map captures, and projects a device-specific random pattern.” Exhibit 35, at 22.

For example, Face ID on the Apple iPhone 17 outputs multiple images comprising light reflected from the scene in order to detect a face in the images.

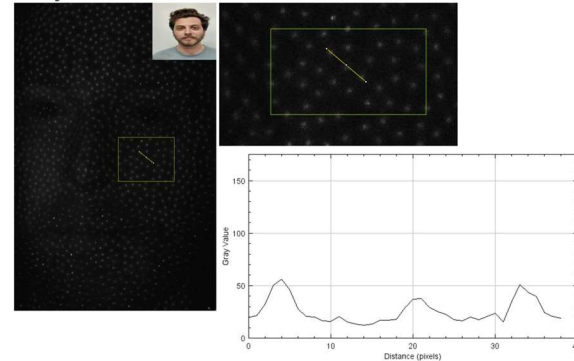
Subject01 Covered area: none – natural Face



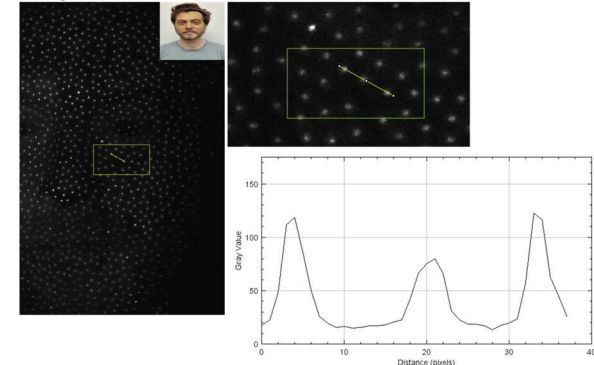
Subject01 Covered area: entire face



Subject02 Covered area: none – natural Face



Subject02 Covered area: entire face



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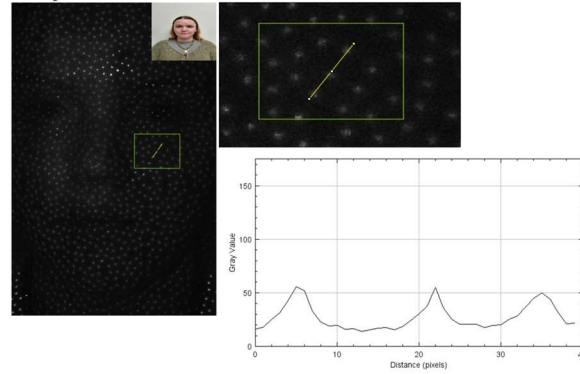
	Exhibit 47 ¶¶ 23-31.
[1g] analyze the reflective light pattern to determine whether the detected face includes depth information consistent with a three-dimensional object;	The Accused Apple Products comprise a processor configured to analyze the reflective light pattern to determine whether the detected face includes depth information consistent with a three-dimensional object. “With a simple glance, Face ID securely unlocks supported Apple devices. It provides intuitive and secure authentication enabled by the TrueDepth camera system, which uses advanced technologies to accurately map the geometry of a user’s face. Face ID uses neural networks for determining attention, matching, and antispoofing, so a user can unlock their phone with a glance, even with a mask on when using supported devices.” Exhibit 35, at 22. “After the TrueDepth camera confirms the presence of an attentive face, it projects and reads thousands of infrared dots to form a depth map of the face along with a 2D infrared image. This data is used to create a sequence of 2D images and depth maps, which are digitally signed and sent to the Secure Enclave. To counter both digital and physical spoofs, the TrueDepth camera randomizes the sequence of 2D images and depth map captures, and projects a device-specific random pattern. A portion of the Secure Neural Engine—protected within the Secure Enclave—transforms this data into a mathematical representation and compares that representation to the enrolled facial data. This enrolled facial data is itself a mathematical representation of the user’s face captured across a variety of poses.” Exhibit 35, at 22. “Apple developed the facial matching neural networks using over a billion images, including infrared (IR) and depth images collected in studies conducted with the participants’ informed consent.” Exhibit 35, at 29. “An additional neural network—that’s trained to spot and resist spoofing—defends against attempts to unlock the device with photos or masks.” Exhibit 35, at 29.

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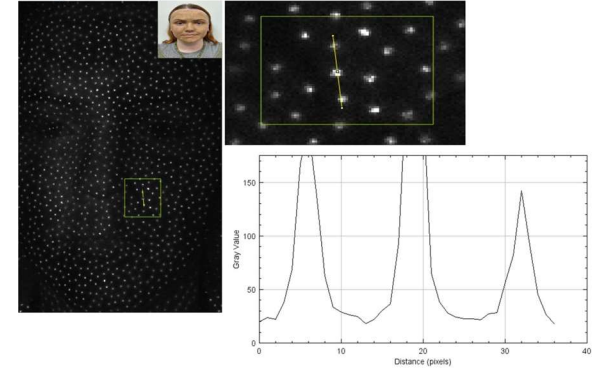
[1h] determine a material classification of the detected face based on a comparison of light intensity distributions in the second image and in a reference image;	The Accused Apple Products comprise a processor configured to determine a material classification of the detected face based on a comparison of light intensity distributions in the second image and in a reference image. “With a simple glance, Face ID securely unlocks supported Apple devices. It provides intuitive and secure authentication enabled by the TrueDepth camera system, which uses advanced technologies to accurately map the geometry of a user’s face. Face ID uses neural networks for determining attention, matching, and antispoofing, so a user can unlock their phone with a glance, even with a mask on when using supported devices.” Exhibit 35, at 22. “After the TrueDepth camera confirms the presence of an attentive face, it projects and reads thousands of infrared dots to form a depth map of the face along with a 2D infrared image. This data is used to create a sequence of 2D images and depth maps, which are digitally signed and sent to the Secure Enclave. To counter both digital and physical spoofs, the TrueDepth camera randomizes the sequence of 2D images and depth map captures, and projects a device-specific random pattern.” Exhibit 35, at 22. “Facial matching is performed within the Secure Enclave using neural networks trained specifically for that purpose. Apple developed the facial matching neural networks using over a billion images, including infrared (IR) and depth images collected in studies conducted with the participants’ informed consent. Apple then worked with participants from around the world to include a representative group of people accounting for gender, age, ethnicity, and other factors. The studies were augmented as needed to provide a high degree of accuracy for a diverse range of users.” Exhibit 35, at 29. “An additional neural network—that’s trained to spot and resist spoofing—defends against attempts to unlock the device with photos or masks.” Exhibit 35, at 29. For example, Face ID on the Apple iPhone 17 succeeds when imaging faces of certain materials but fails when imaging faces of other materials, based on reference data.
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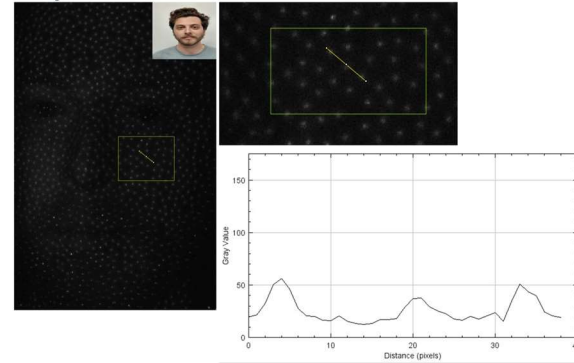
Subject01 Covered area: none – natural Face



Subject01 Covered area: entire face



Subject02 Covered area: none – natural Face



Subject02 Covered area: entire face

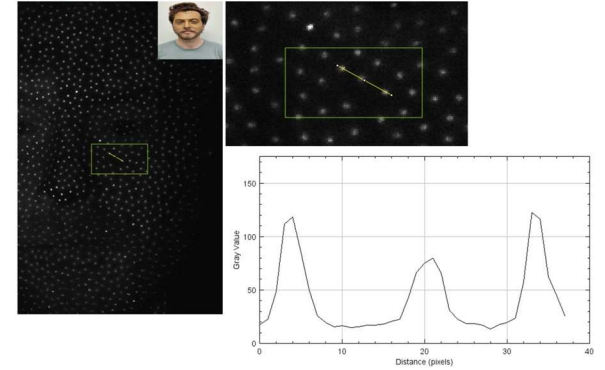


Exhibit 47 ¶¶ 23-31.

Face ID further uses neural networks to compare the given regions to corresponding regions in reference image data.

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Face ID security

With a simple glance, Face ID securely unlocks supported Apple devices. It provides intuitive and secure authentication enabled by the TrueDepth camera system, which uses advanced technologies to accurately map the geometry of a user’s face. Face ID uses neural networks for determining attention, matching, and antispoofing, so a user can unlock their phone with a glance, even with a mask on when using supported devices. Face ID automatically adapts to changes in appearance, and carefully safeguards the privacy and security of a user’s biometric data.

Face ID is designed to confirm user attention, provide robust authentication with a low false-match rate, and mitigate both digital and physical spoofing.

The TrueDepth camera automatically looks for the user’s face when the user wakes an Apple device that features Face ID (by raising it or tapping the screen), as well as when those devices attempt to authenticate the user to display an incoming notification or when a supported app requests Face ID authentication. When a face is detected, Face ID confirms attention and intent to unlock by detecting that the user’s eyes are open and their attention is directed at their device; for accessibility, the Face ID attention check is disabled when VoiceOver is activated and, if required, can be disabled separately. Attention detection is always required when using Face ID with a mask.

After the TrueDepth camera confirms the presence of an attentive face, it projects and reads thousands of infrared dots to form a depth map of the face along with a 2D infrared image. This data is used to create a sequence of 2D images and depth maps, which are digitally signed and sent to the Secure Enclave. To counter both digital and physical spoofs, the TrueDepth camera randomizes the sequence of 2D images and depth map captures, and projects a device-specific random pattern. A portion of the Secure Neural Engine—protected within the Secure Enclave—transforms this data into a mathematical representation and compares that representation to the enrolled facial data. This enrolled facial data is itself a mathematical representation of the user’s face captured across a variety of poses.

Exhibit 35, at 22.

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Facial matching security

Facial matching is performed within the Secure Enclave using neural networks trained specifically for that purpose. Apple developed the facial matching neural networks using over a billion images, including infrared (IR) and depth images collected in studies conducted with the participants’ informed consent. Apple then worked with participants from around the world to include a representative group of people accounting for gender, age, ethnicity, and other factors. The studies were augmented as needed to provide a high degree of accuracy for a diverse range of users. Face ID is designed to work with hats, scarves, eyeglasses, contact lenses, and many types of sunglasses. Face ID also supports unlocking with a mask on iPhone devices starting with iPhone 12 and iOS 15.4 or later. Furthermore, it’s designed to work indoors, outdoors, and even in total darkness. An additional neural network—that’s trained to spot and resist spoofing—defends against attempts to unlock the device with photos or masks. Face ID data, including mathematical representations of a user’s face, is encrypted and available only to the Secure Enclave. This data never leaves the device. It’s not sent to Apple, nor is it included in device backups. The following Face ID data is saved, encrypted only for use by the Secure Enclave, during normal operation:

- The mathematical representations of a user’s face calculated during enrollment
- The mathematical representations of a user’s face calculated during some unlock attempts if Face ID deems them useful to augment future matching

Face images captured during normal operation aren’t saved but are instead immediately discarded after the mathematical representation is calculated for either enrollment in Face ID or comparison with enrolled Face ID data.

Exhibit 35, at 29.

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	Advanced technologies The technology that enables Face ID is some of the most advanced hardware and software that we’ve ever created. The TrueDepth camera captures accurate face data by projecting and analyzing thousands of invisible dots to create a depth map of your face and also captures an infrared image of your face. A portion of your device's neural engine — protected within the Secure Enclave — transforms the depth map and infrared image into a mathematical representation and compares that representation to the enrolled facial data. Exhibit 36. Face ID matches against depth information, which isn’t found in print or 2D digital photographs. It's designed to protect against spoofing by masks or other techniques through the use of sophisticated anti-spoofing neural networks. Face ID is even attention-aware, and Face ID with a mask will always confirm attention. Face ID recognizes if your eyes are open and your attention is directed towards the device. This makes it more difficult for someone to unlock your device without your knowledge (such as when you are sleeping). Exhibit 36.
[1i] authenticate the detected face based on the depth information and the material classification.	The Accused Apple Products comprise a processor configured to authenticate the detected face based on the depth information and the material classification. “With a simple glance, Face ID securely unlocks supported Apple devices. It provides intuitive and secure authentication enabled by the TrueDepth camera system, which uses advanced technologies to accurately map the geometry of a user’s face. Face ID uses neural networks for determining attention, matching, and antispoofing, so a user can unlock their phone with a glance, even with a mask on when using supported devices.” Exhibit 35, at 22. “After the TrueDepth camera confirms the presence of an attentive face, it projects and

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	reads thousands of infrared dots to form a depth map of the face along with a 2D infrared image. This data is used to create a sequence of 2D images and depth maps, which are digitally signed and sent to the Secure Enclave. To counter both digital and physical spoofs, the TrueDepth camera randomizes the sequence of 2D images and depth map captures, and projects a device-specific random pattern. A portion of the Secure Neural Engine—protected within the Secure Enclave—transforms this data into a mathematical representation and compares that representation to the enrolled facial data. This enrolled facial data is itself a mathematical representation of the user’s face captured across a variety of poses.” Exhibit 35, at 22. “Apple developed the facial matching neural networks using over a billion images, including infrared (IR) and depth images collected in studies conducted with the participants’ informed consent.” Exhibit 35, at 29. “An additional neural network—that’s trained to spot and resist spoofing—defends against attempts to unlock the device with photos or masks.” Exhibit 35, at 29.
Claim 17	Accused Apple Products
[17pre] A system comprising:	The Accused Apple Products comprise a system. <i>See supra</i> analysis for limitation [1pre].
[17a] a near infrared (NIR) camera configured to capture a scene, the NIR camera comprising a CMOS sensor;	The Accused Apple Products comprise a near infrared (NIR) camera configured to capture a scene, the NIR camera comprising a CMOS sensor. <i>See supra</i> analysis for limitation [1a].
[17b] an illumination unit comprising:	The Accused Apple Products comprise an illumination unit. <i>See supra</i> analysis for limitation [1b].
[17c] at least one of a first LED and a first laser configured to	The Accused Apple Products comprise an illumination unit comprising at least one of a first LED and a first laser configured to project NIR light in a light pattern toward the scene.

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project NIR light in a light pattern toward the scene; and	<i>See supra</i> analysis for limitation [1b].
[17d] at least one of a second LED and a second laser configured to project an NIR light beam toward the scene;	The Accused Apple Products comprise an illumination unit comprising at least one of a second LED and a second laser configured to project an NIR light beam toward the scene. <i>See supra</i> analysis for limitation [1b].
[17e] at least one optical element, wherein the NIR light passes through the at least one optical element;	The Accused Apple Products comprise at least one optical element, wherein the NIR light passes through the at least one optical element. <i>See supra</i> analysis for limitation [1c].
[17f] a processor configured to:	The Accused Apple Products comprise a processor. <i>See supra</i> analysis for limitation [1d].
[17g] output a first image comprising light reflected from the scene in response to the projection of the NIR light beam;	The Accused Apple Products comprise a processor configured to output a first image comprising light reflected from the scene in response to the projection of the NIR light beam. <i>See supra</i> analysis for limitation [1e].
[17h] detect a face in the first image;	The Accused Apple Products comprise a processor configured to detect a face in the first image. <i>See supra</i> analysis for limitation [1e].
[17i] output a second image comprising a reflective light pattern reflected from the scene in response to the projection of the NIR light in the light pattern;	The Accused Apple Products comprise a processor configured to output a second image comprising a reflective light pattern reflected from the scene in response to the projection of the NIR light in the light pattern. <i>See supra</i> analysis for limitation [1f].
[17j] analyze the reflective light pattern to determine whether the detected face includes depth	The Accused Apple Products comprise a processor configured to analyze the reflective light pattern to determine whether the detected face includes depth information consistent with a three-dimensional object.

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information consistent with a three-dimensional object;	<i>See supra</i> analysis for limitation [1g].
[17k] determine a material classification of the detected face based on a comparison of light intensity distributions in the second image and in a reference image;	The Accused Apple Products comprise a processor configured to determine a material classification of the detected face based on a comparison of light intensity distributions in the second image and in a reference image. <i>See supra</i> analysis for limitation [1h].
[17l] authenticate the detected face based on the depth information and the material classification.	The Accused Apple Products comprise a processor configured to authenticate the detected face based on the depth information and the material classification. <i>See supra</i> analysis for limitation [1i].